

LAB REPORT
FOR
LAB - 2
EEC 556: SOFTWARE DEFINED RADIO
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LAB 2: BASEBAND QAM MODEM

PART 1: MODULATION AND DETECTION

The motto of this lab is to develop a baseband digital modem capable of using BPSK and QPSK modulation. Key concepts covered in this lab include constellation mapping, oversampling, pulse shaping, matched filtering, maximum likelihood detection, and error probability calculation.

The sequence of bits $\{b[n]\}$ is passed to a radio's physical layer by higher layers, which symbol-maps it to provide a sequence of symbols from one of the sets of constellations $\mathcal{C}$. In real-world implementations, M , the number of symbols, is given by $(M = 2^b)$, where b is the number of bits per symbol. To facilitate implementation, normalization of the constellation to have "unit energy," so that the symbol energy is fixed, is straightforward. The symbols are then pulse-shaped and scaled by the symbol energy to create a complex baseband signal. The shape of the pulse is represented by the function $(g_{\{tx\}}(t))$, and to preserve energy, the function is normalized over the length of the entire waveform. The period of the symbol is T , with the symbol rate being $(1/T)$ and the bit rate being (b/T) .

The resulting generated signal is then upconverted by Analog equipment to be transmitted. In practical uses, pulse shaping is done in discrete time. When transmitted, the signal goes through the wireless channel and gets corrupted by noise and other channel distortions. However, in this lab, the transmitted signal is assumed to be degraded by only Additive White Gaussian Noise (AWGN). The optimum receiver design for this type of transmission is matched filtering, symbol rate sampling, and reception symbol detection. The received signal is sampled at a rate satisfying the Nyquist criterion at the receiver end. The signal is then filtered by a matched filter, whose filter impulse response $(g_{\{rx\}}(n))$ is the time-reversal of the transmit pulse shape, i.e., $(g_{\{rx\}}(t) = g_{\{tx\}}(-t))$. After digital filtering, the signal is downsampled by M to produce the symbol rate sampled signal, $(y[n])$.

The lab 2.1 is divided into two parts: Modulation and Detection.

Modulation :

After downloading lab files applying changes to student modulate.vi for both BPSK AND QPSK as shown in figure 1 and figure 2.

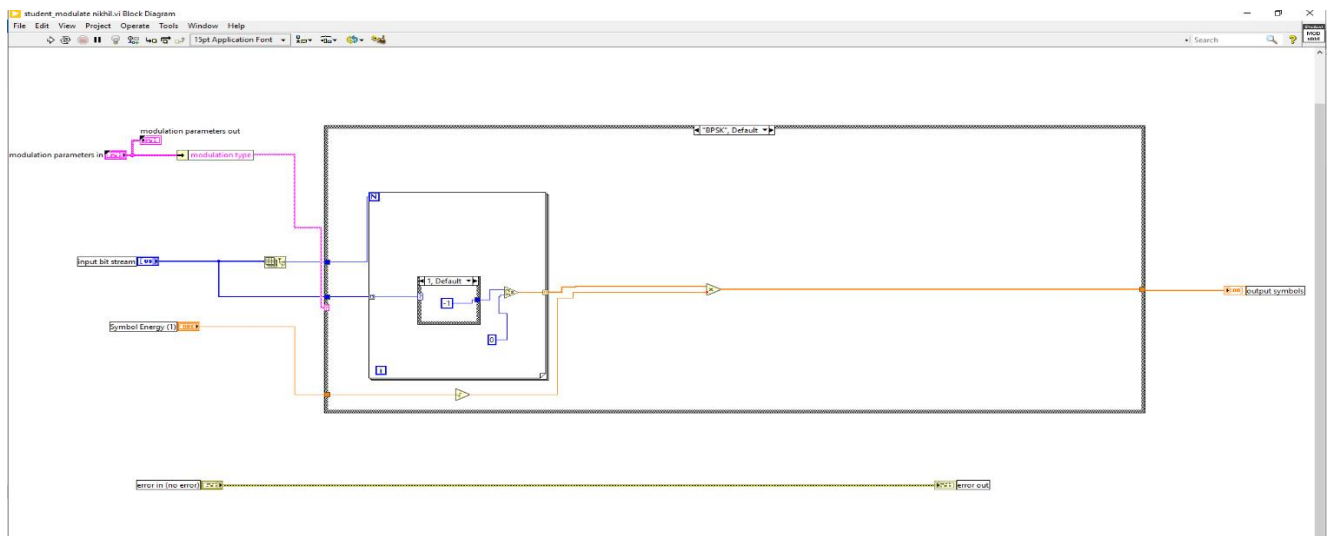


figure 1: student_modulate.nikhil.vi (BPSK block diagram)

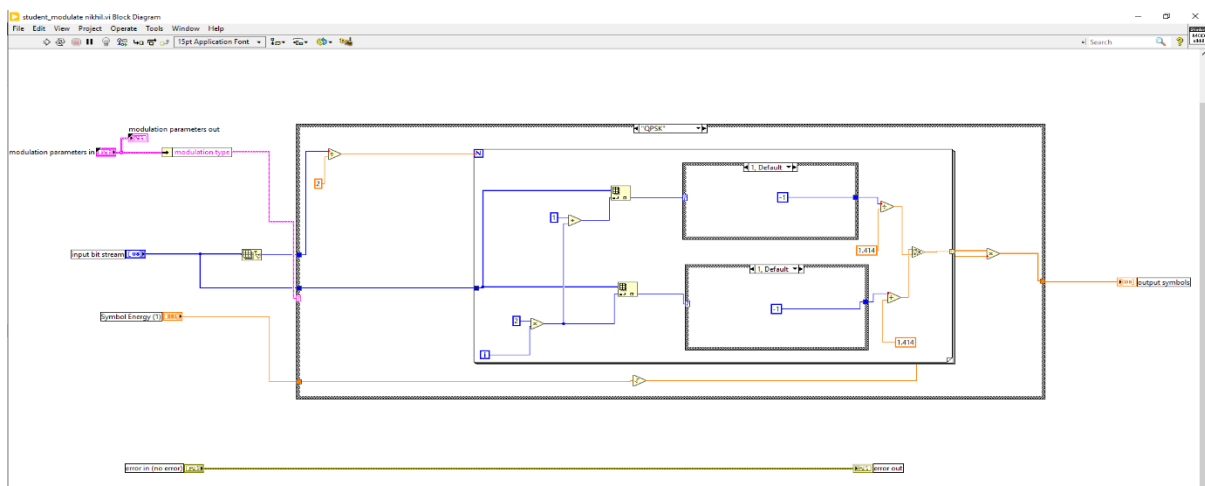


Figure 2: student_modulate.nikhil.vi (QPSK block diagram)

After saving this files with changes we have observed modulation front panel results as shown in Figure 3 & Figure 4.

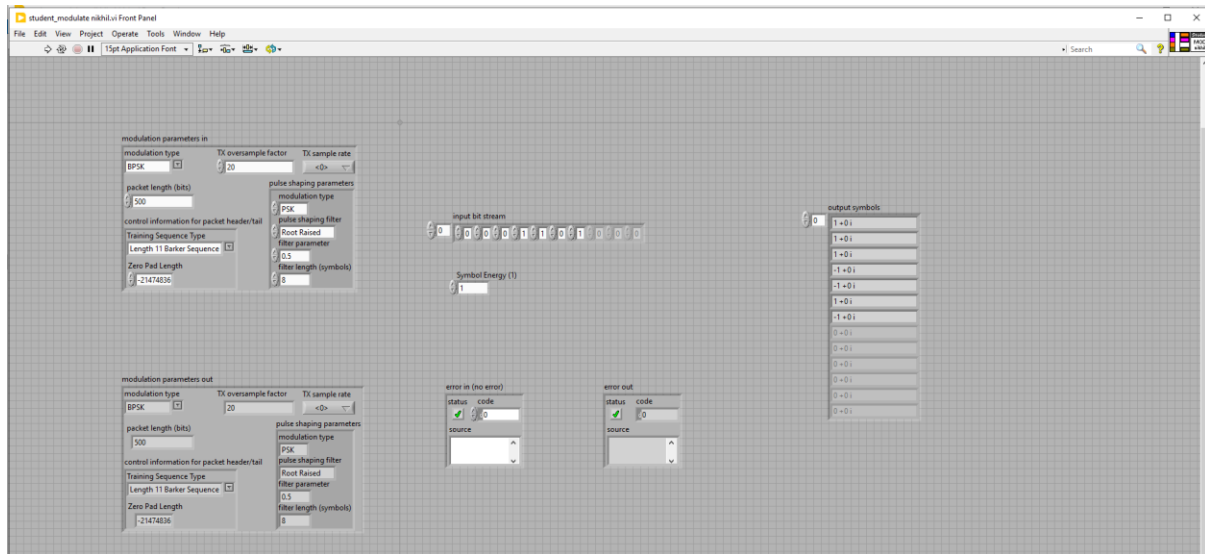


Figure 3: student_modulate nikhil.vi (BPSK front panel)

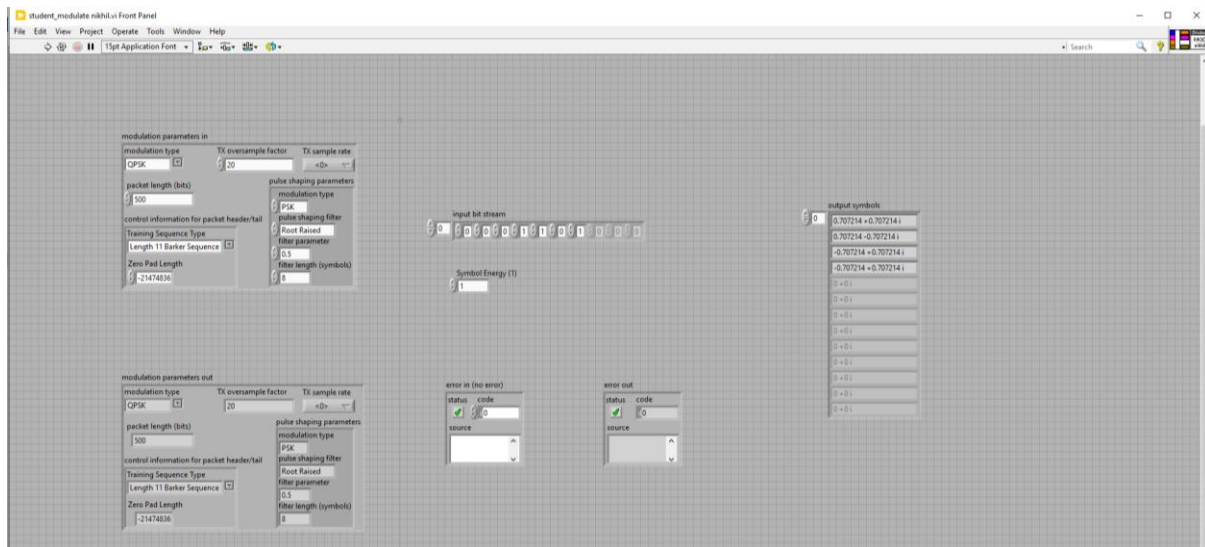


Figure 4 : student_modulate nikhil.vi (QPSK front panel)

Detection:

After modulation, we have done decode part as shown in Figure 5 & Figure 6.

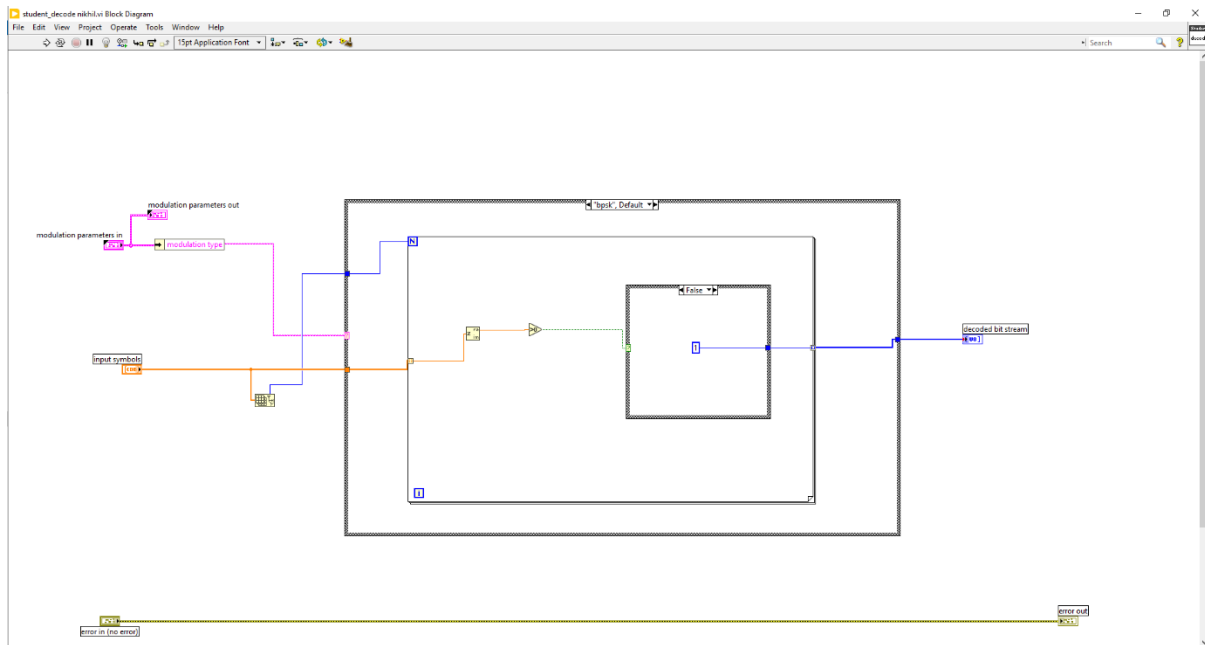


Figure 5: student_decode_nikhil.vi(BPSK Block diagram)

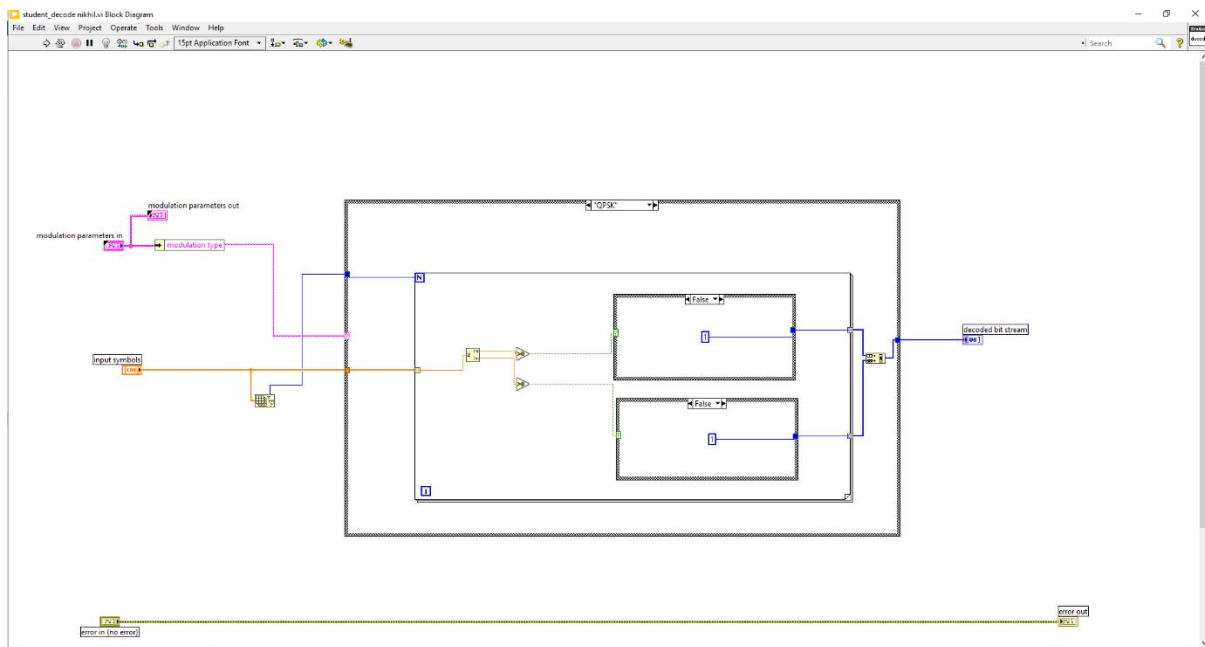


Figure 6: student_decode_nikhil.vi(QPSK block diagram)

The image displays a MATLAB/Simulink workspace with the following components:

- Script Editor:** Contains the `student_decode.m` script, which is organized into three main sections:
 - modulation parameters in:** Defines RX oversample factor (20), RX Sample Rate (4M), and various modulation parameters (PSK, Root Raised, filter length, etc.).
 - modulation parameters out:** Defines similar parameters for the output.
 - channel estimation/qualifier:** Defines channel estimation and equalization parameters.
- Block Diagram:** A Simulink model titled `student_decode` showing the data flow from `input symbols` through a `modulation` block to a `demodulation` block, which then outputs `decoded bit stream`. The diagram also includes `error in (no error)` and `error out` blocks for error detection.

[illegible]

After saving this files we have replaced the files modulate and decode in transmitter and receiver respectively as shown in Figure 9 & Figure 10.

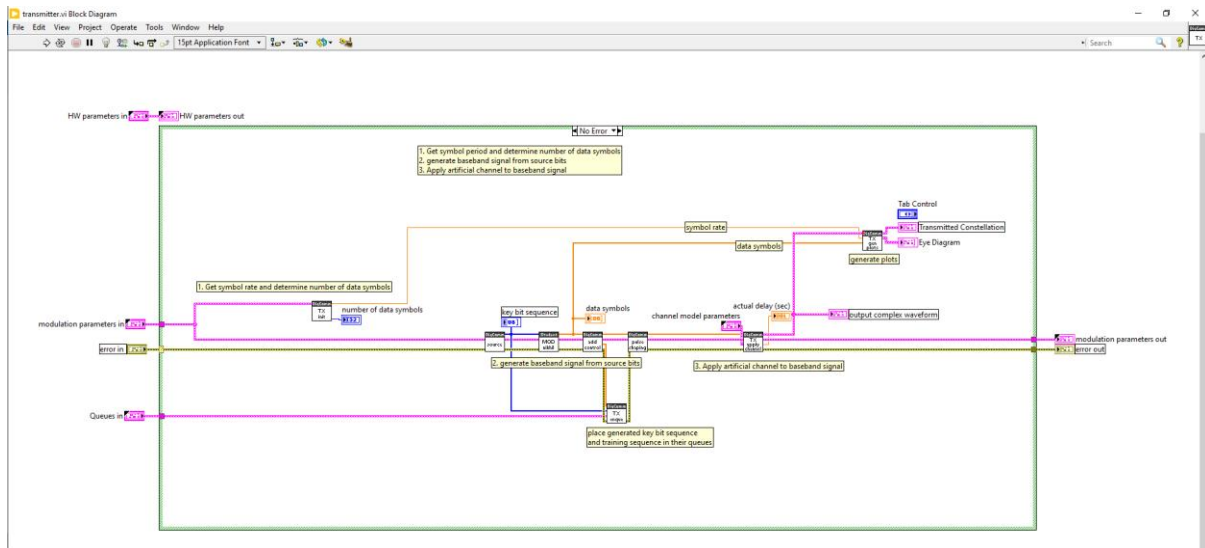


Figure 9: Transmitter.vi(Block diagram)

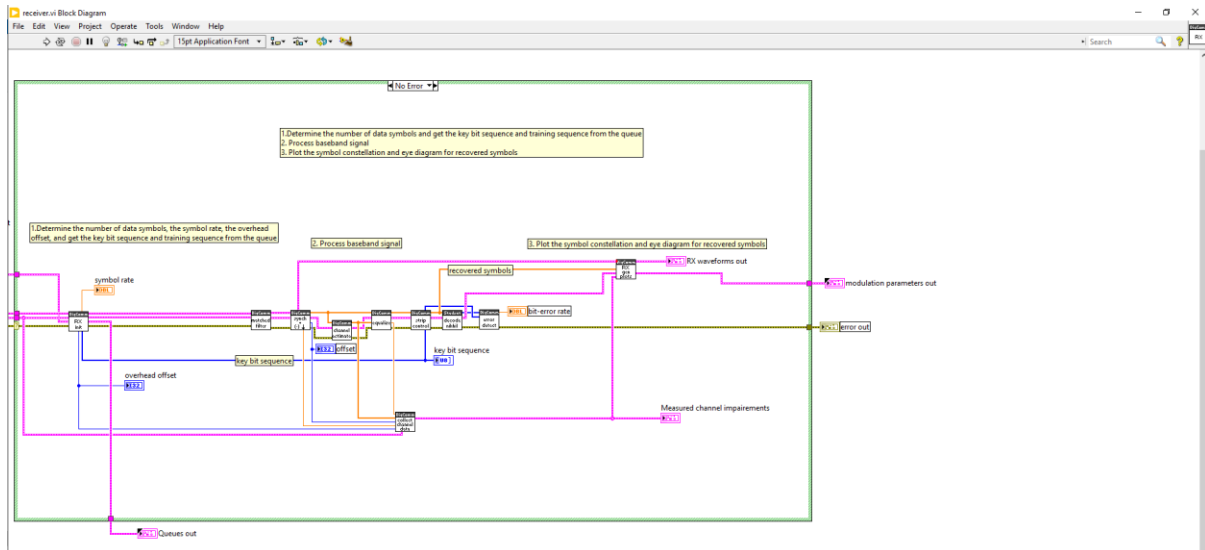


Figure 10: Receiver.vi(Block diagram)

After Replacing both the vi's we have have observed simulator results as shown in Figure 11 & Figure 12.

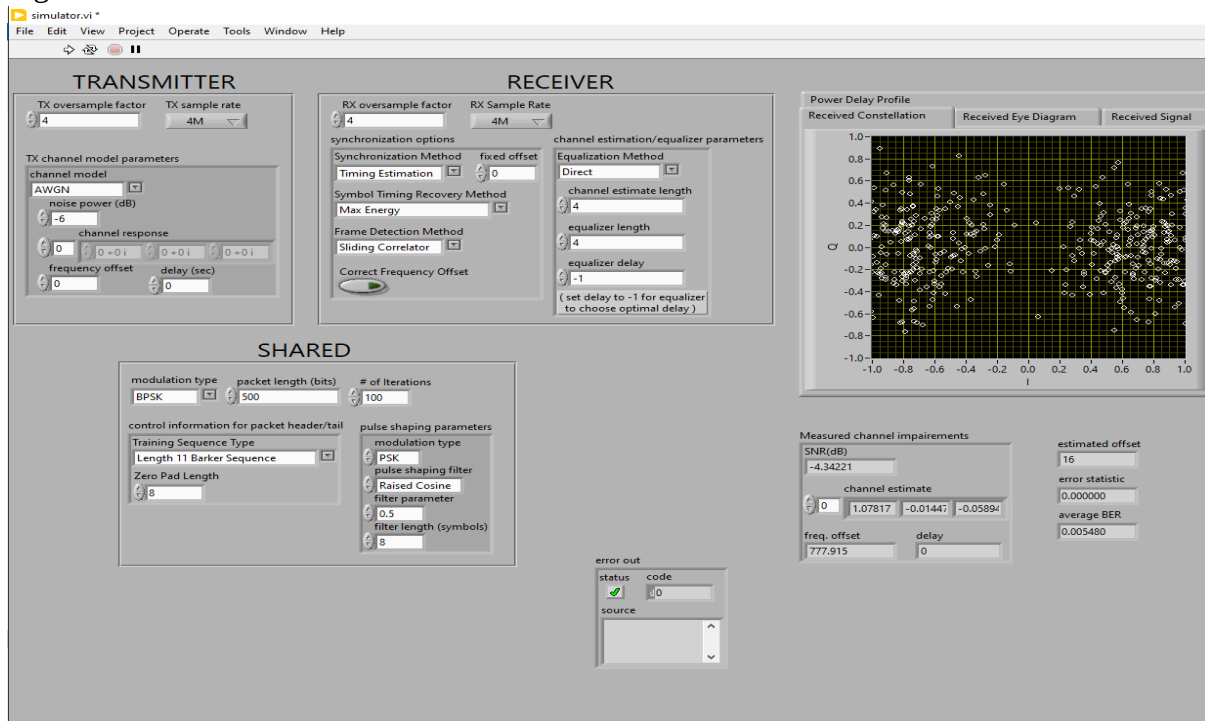


Figure 11: simulator.vi (BPSK)

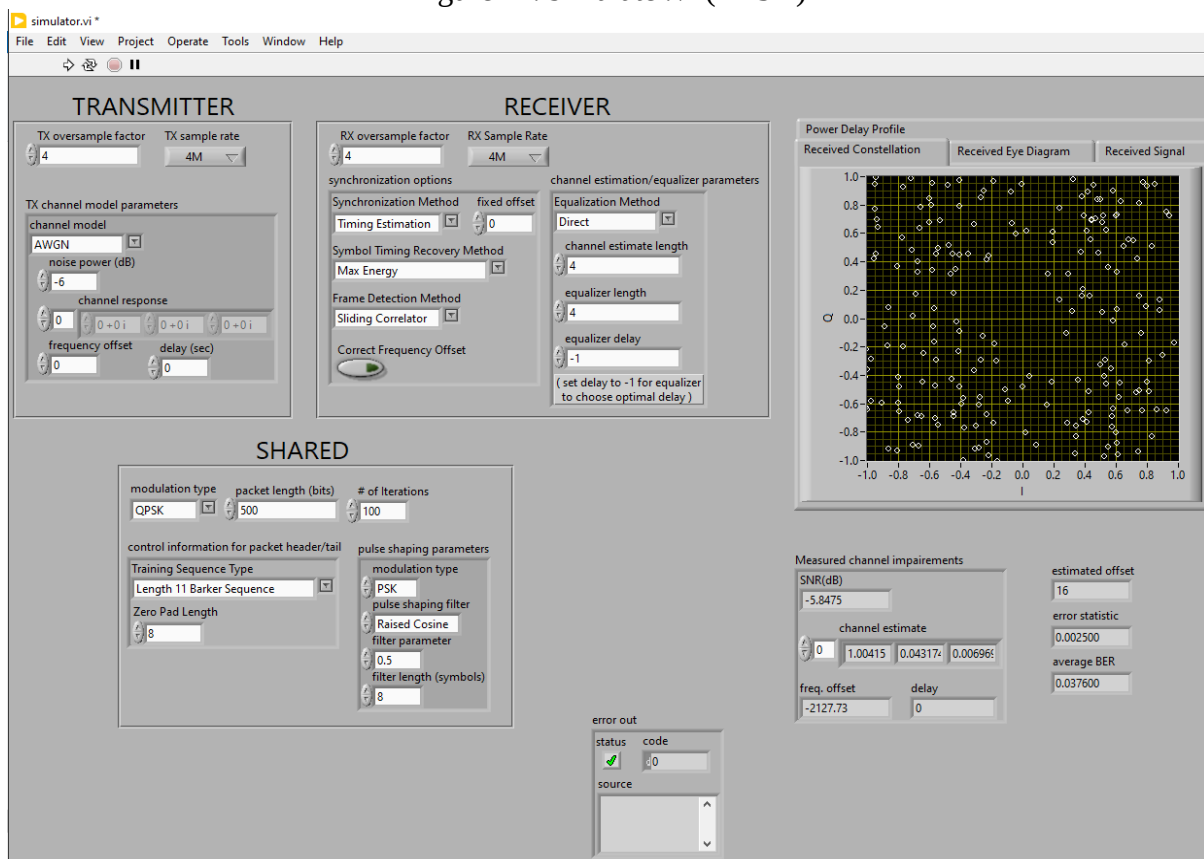


Figure 12: simulator.vi(QPSK)

Next we have connected Ni usrp devices to computers and transmitted and received signals as shown in below figures.

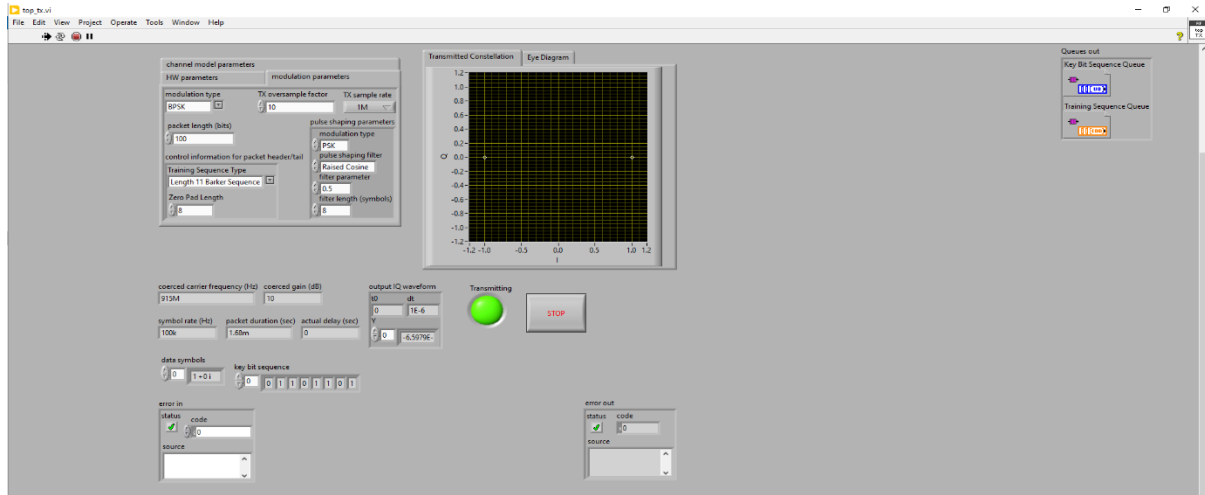


Figure 13:top_tx.vi(BPSK transmitted signal constellation)

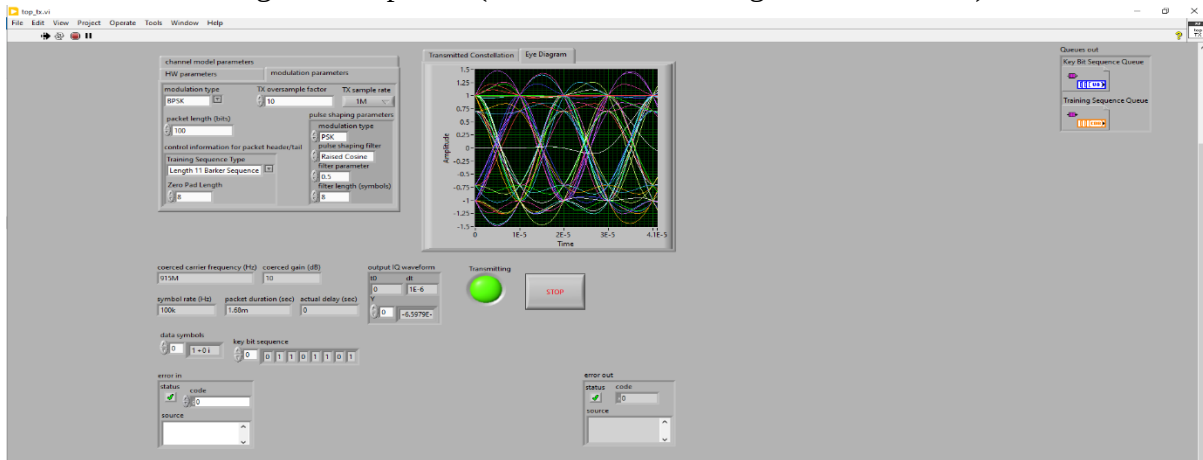


Figure 14:top_tx.vi(BPSK Eye diagram)

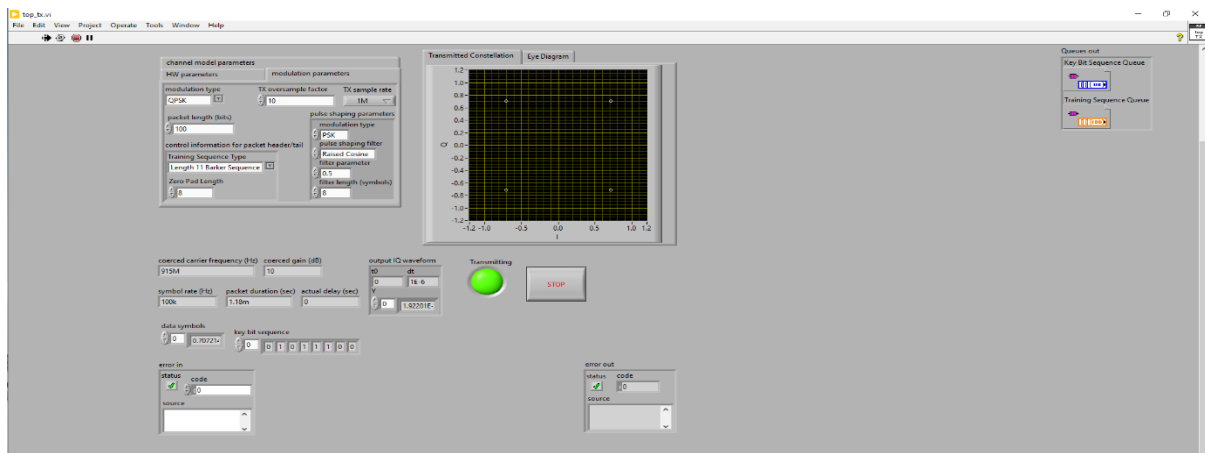


Figure 15:top_tx.vi(QPSK transmitted signal constellation)

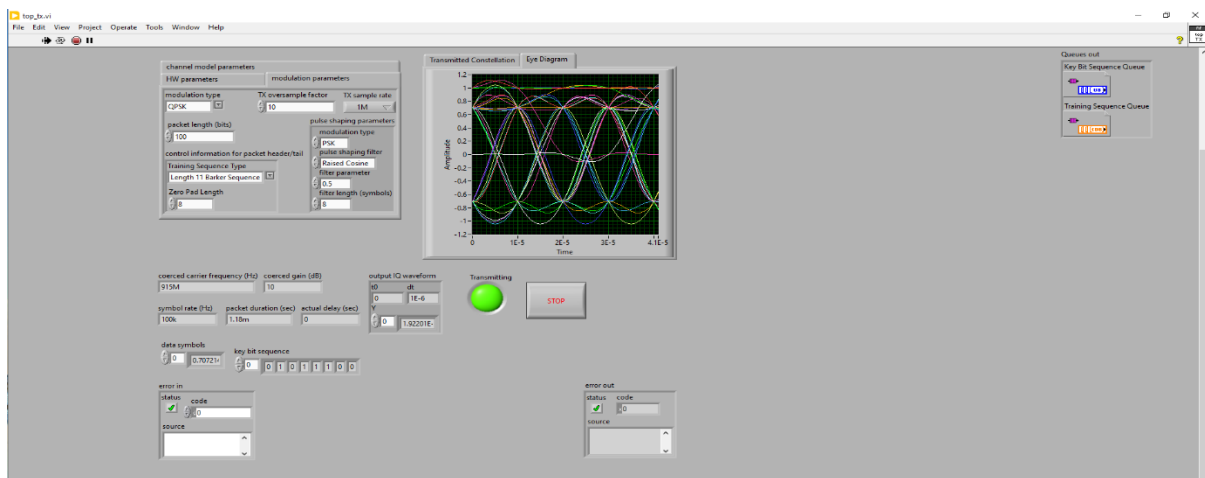


Figure 16:top_tx.vi(QPSK Eye diagram)

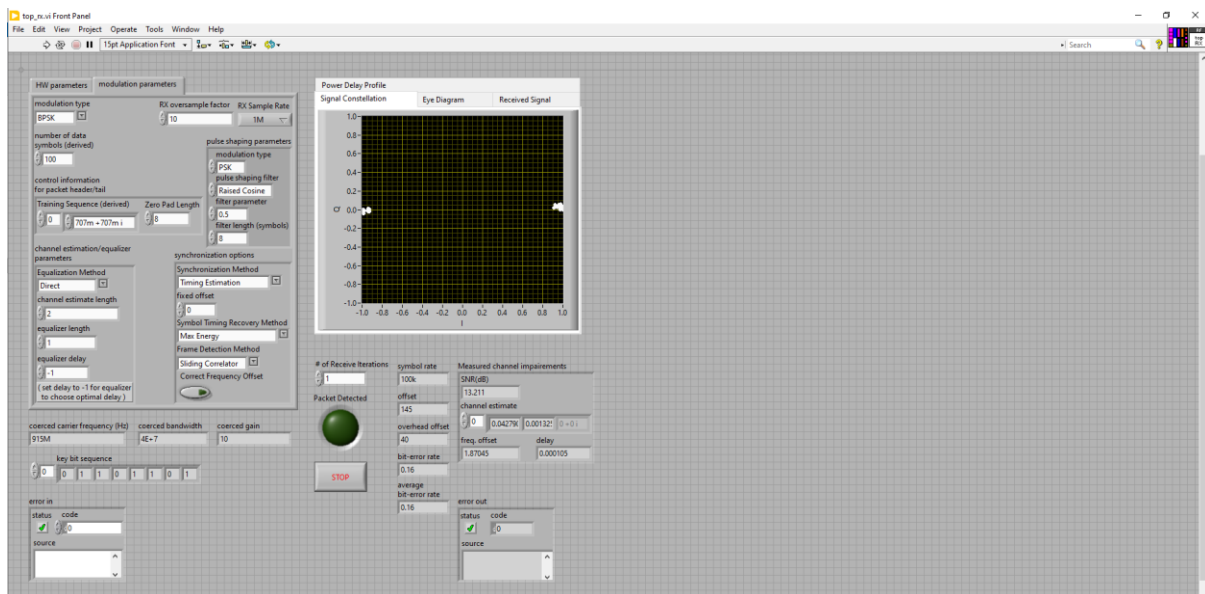


Figure 17:top_rx.vi(BPSK signal constellation)

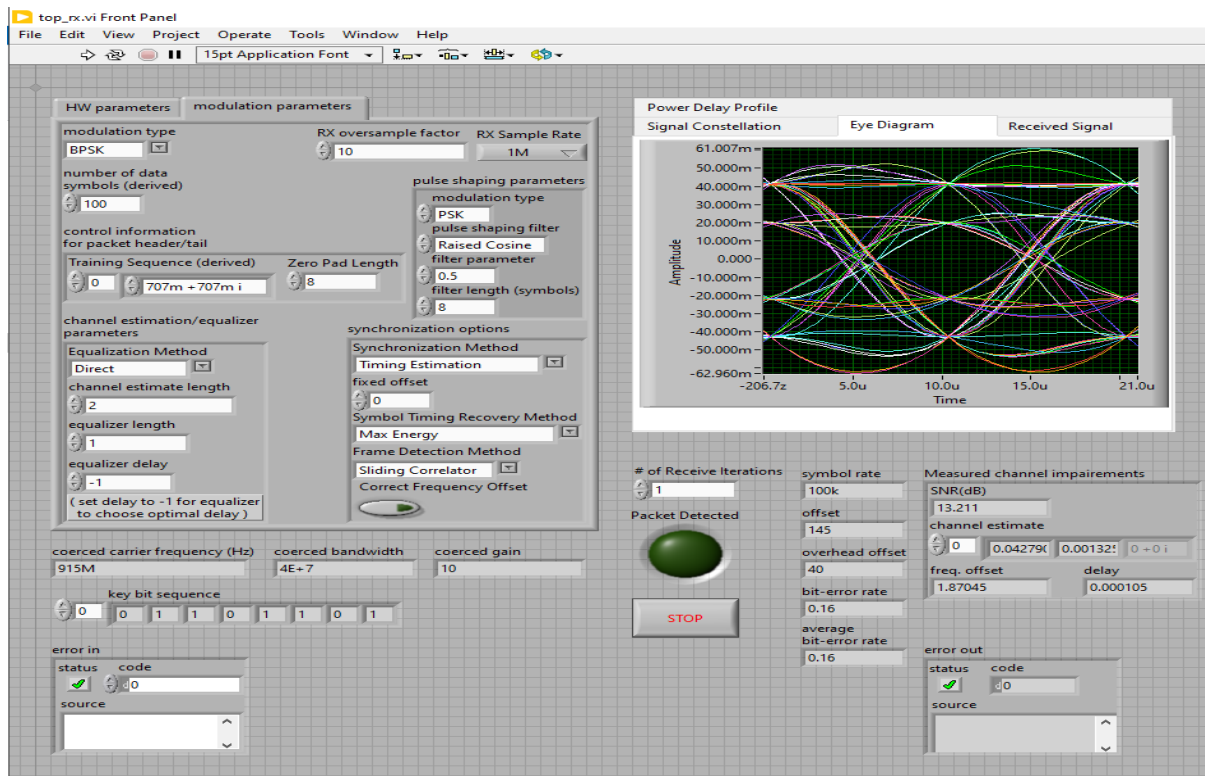


Figure 18:top_rx.vi(BPSK Eye diagram)

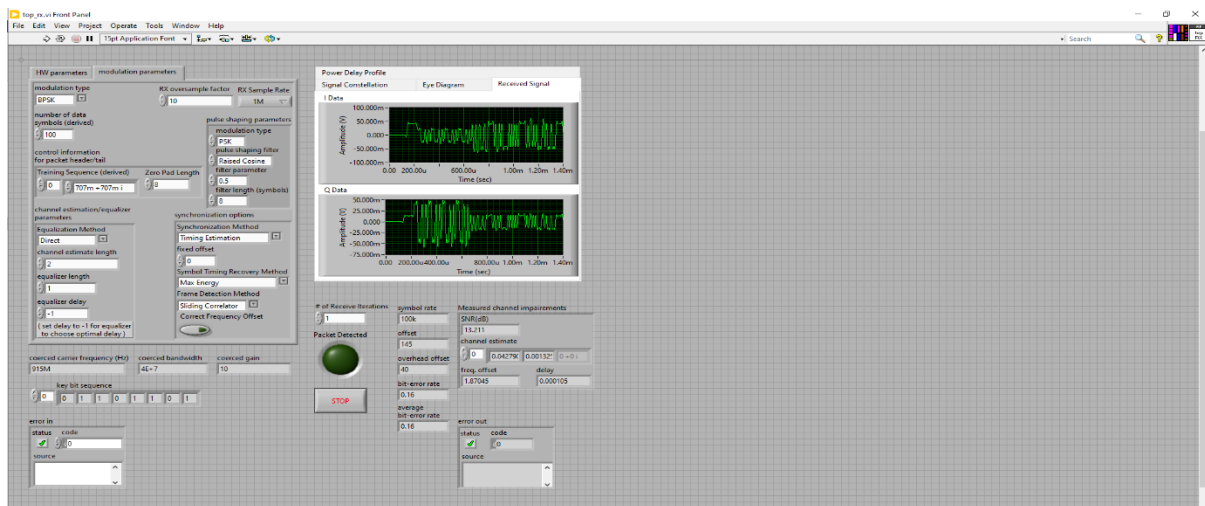


Figure 19:top_rx.vi(BPSK Received signal)

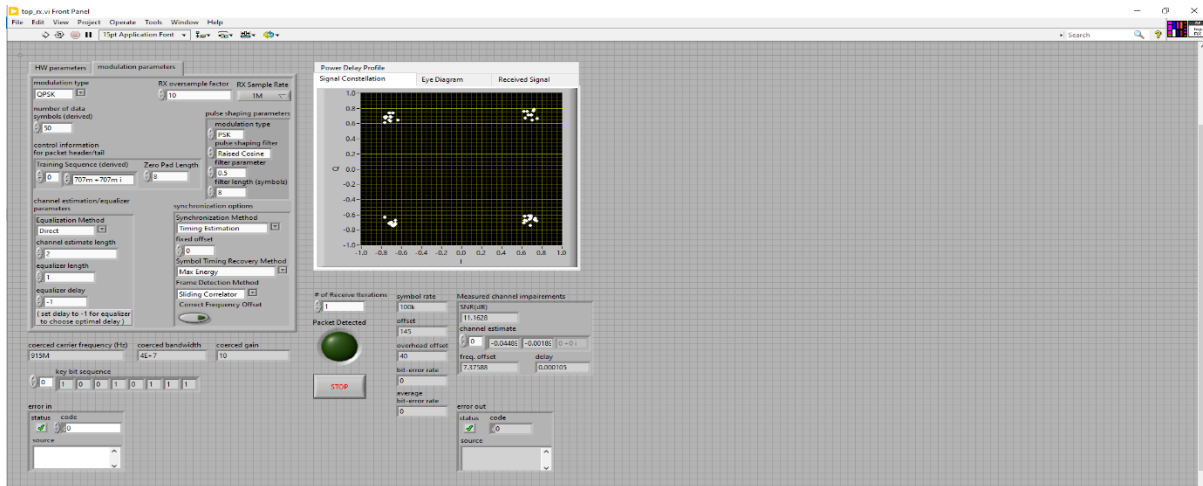


Figure 20:top_rx.vi(QPSK signal constellation)

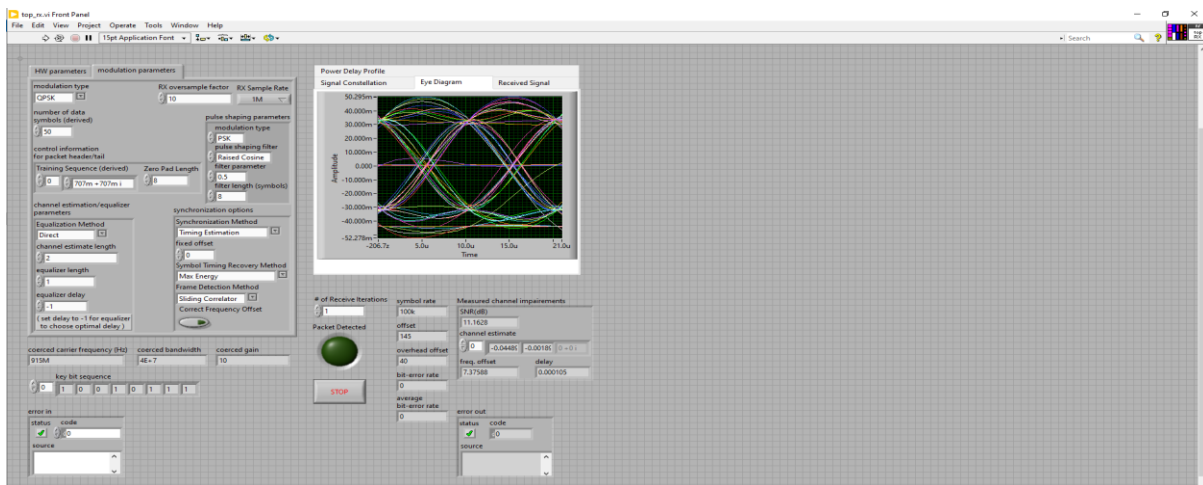


Figure 21:top_rx.vi(QPSK Eye diagram)



Figure 22:top_rx.vi(QPSK Received signal)

Question and answers:

- **PSK is a constant envelope modulation scheme. This means that all symbols have equal energy after modulation. What is the energy of the BPSK and QPSK Modulated symbols in Tables 1(a) and 1(b) respectively?**

Input Bit	symbol
0	1
1	-1

Input Bits	Symbol
00	$1+j$
10	$-1+j$
11	$-1-j$
01	$1-j$

Ans: Energy of BPSK = 1

Energy of QPSK = $\sqrt{2}(|1+j|)$

- **In modulation.vi, What should be the default value for symbol Energy of the Modulated Sequence?**

Ans: Symbol Energy default value is 1

- **What is the name of the queue used to pass the generated bit sequence from top_tx.vi to top_rx.vi.**

Ans: Channel

- **Describe what happens to the receiver constellation as you increase noise power in the top level of the transmitter.**

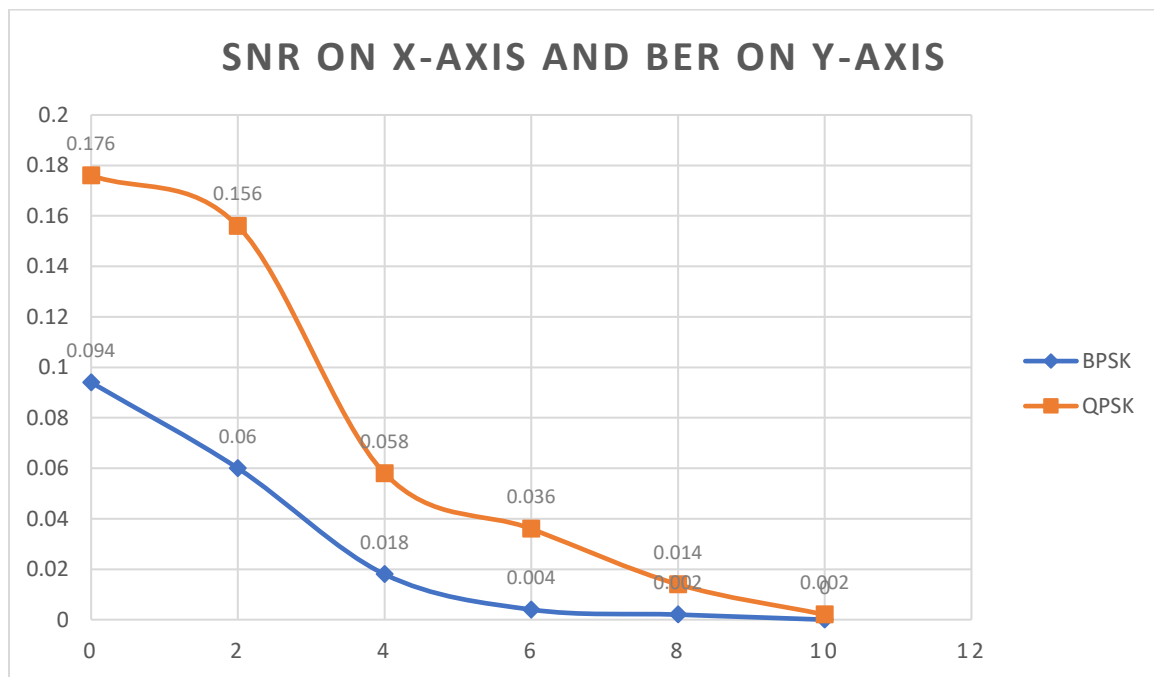
Ans: As the noise power increase:

- significant rise in error probability, received signal may be interpreted a symbol as a different one
 - As high noise result into Low SNR, means Low Quality of signal received.
 - Symbol overlapping increases
- **Based on your observation of the received signal constellation, explain which modulation scheme (BPSK or QPSK) performs better (on-average) noisy channels, and why. Assume that both schemes are using the same average symbol energy.**

Ans: BPSK as BPSK use on 2 bits 0 & 1 so angle will be 180°, however in QPSK there are four symbols located in each quadrant with 90°. In BPSK scheme, due to noise symbol has to travel to very large distance 180° from one symbol to another. While on the other hand in QPSK due to noise symbol has to travel only 90° to be received as an error or wrong symbol. So small amount of noise may cause high error in QPSK scheme. Hence for noisy channels BPSK performs much better than QPSK.

➤ Turn In a Plot which Compare BPSK and QPSK modulation

SNR(db)	0	2	4	6	8	10
BPSK	0.094	0.060	0.018	0.004	0.002	0.000
QPSK	0.176	0.156	0.058	0.036	0.014	0.002



PART 2 :PULSE SHAPING AND MATCHED FILTERING

Digital signals, in their raw form, are not directly suitable for transmission across communication channels. To enable effective transmission, they must first be converted into Analog waveforms that are compatible with the channel's characteristics.

After downloading files we modified with changes in both pulse shaping and matched filtering as shown in figure 23 & figure 24.

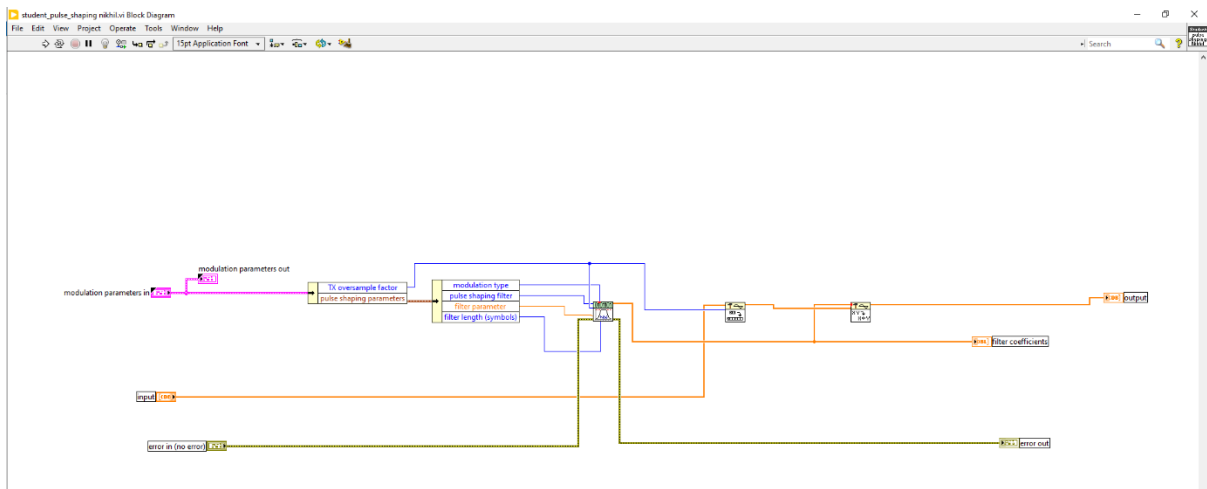


Figure 23:student_pulse_shaping nikhil.vi(Block diagram)

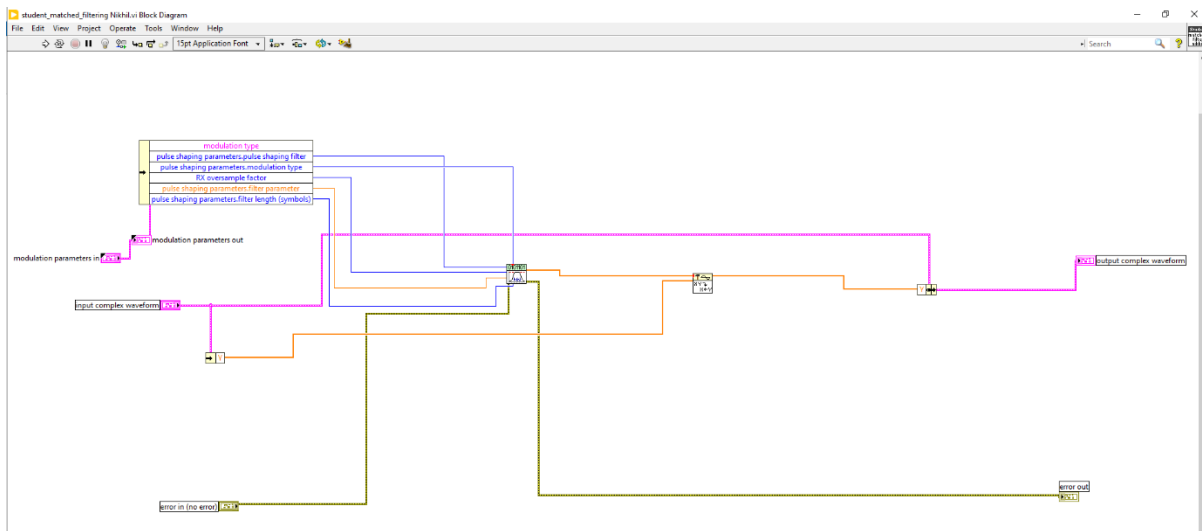


Figure 24:student_matched_filtering Nikhil.vi(Block diagram)

After that we have replaced both pulse shaping and matched filtering in transmitter and receiver in figure 25 & figure 26.

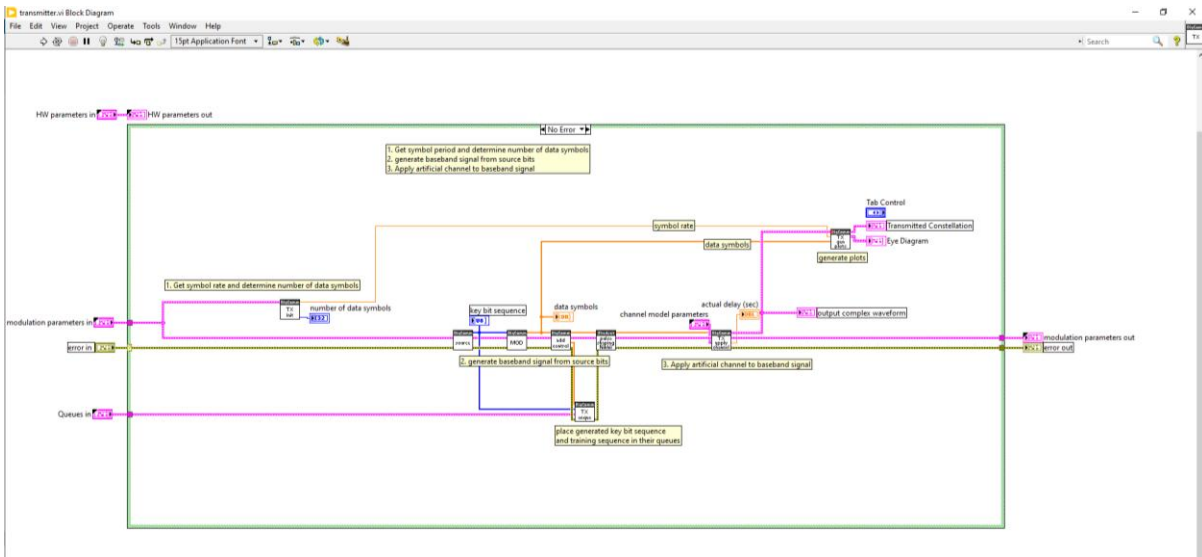


Figure 25:Transmitter.vi(replaced pulse shaping)

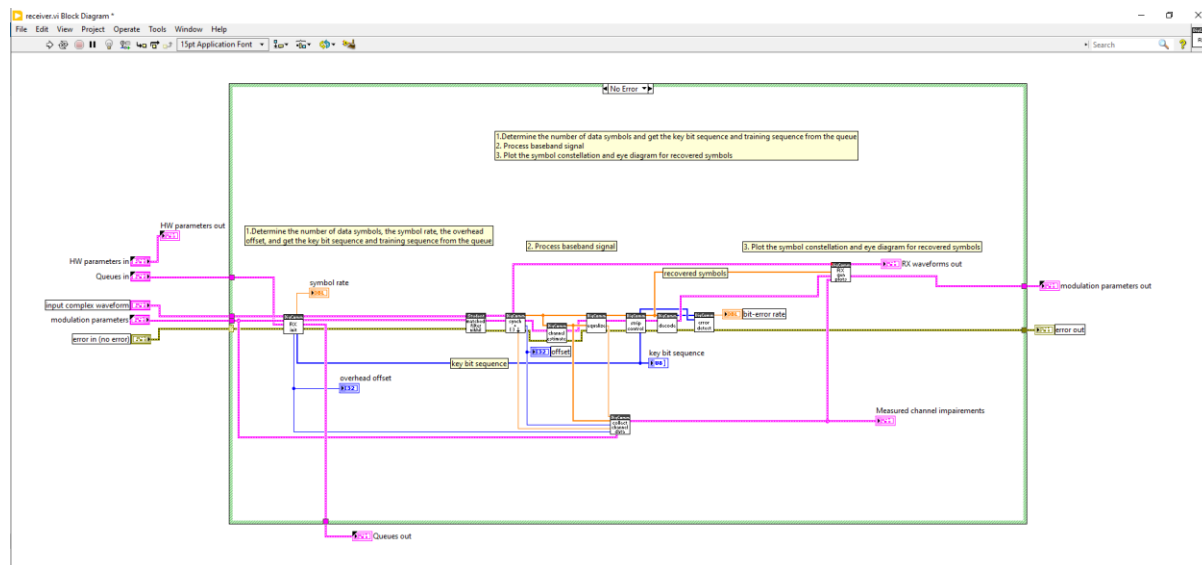


Figure 26:Receiver.vi(replaced matched filtering)

After this we have observed result in simulator.vi as shown in figure 27.

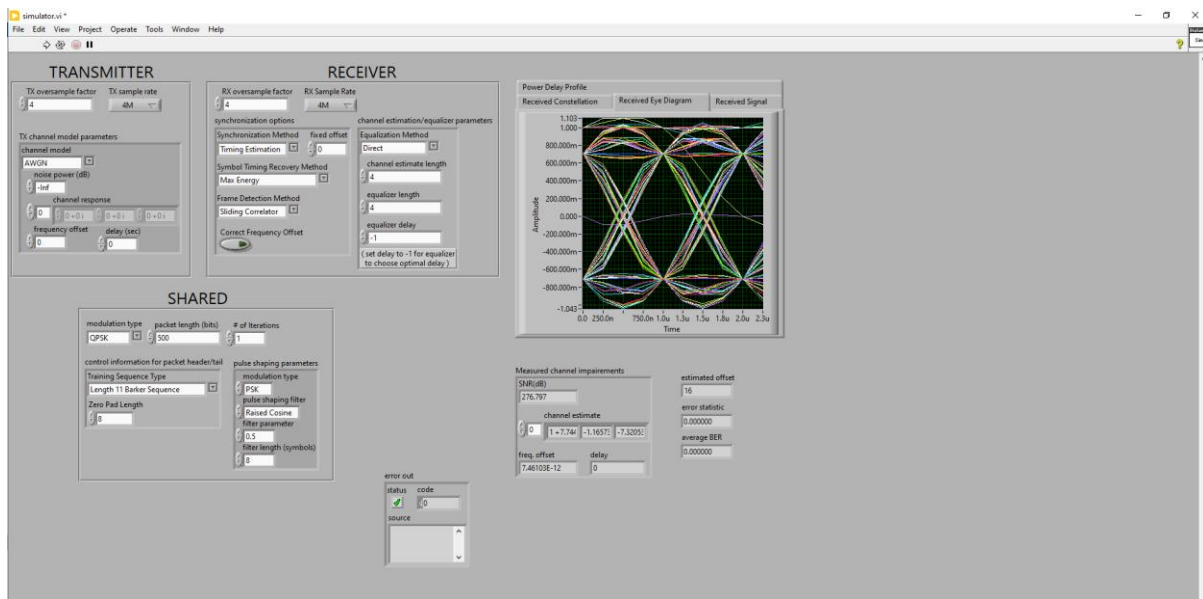


Figure 27: Simulator.vi

We have connecting NI USRP devices to computer we have transmitted and received signal and observed results , as shown in figure 28 & figure 29.

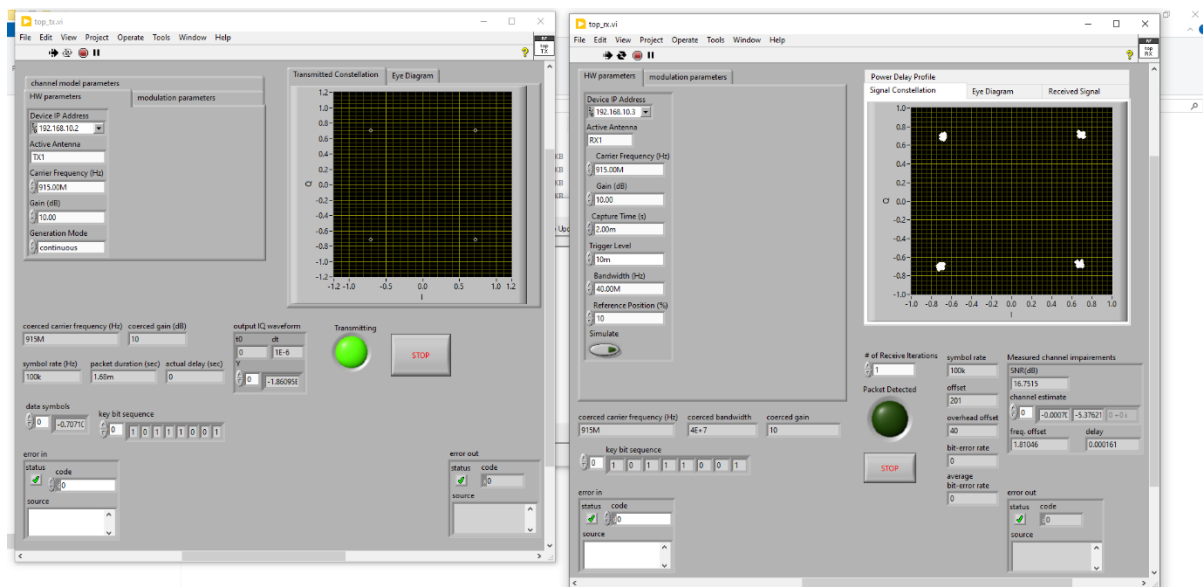


Figure 28:top_tx.vi & top_rx.vi(signal constellation)

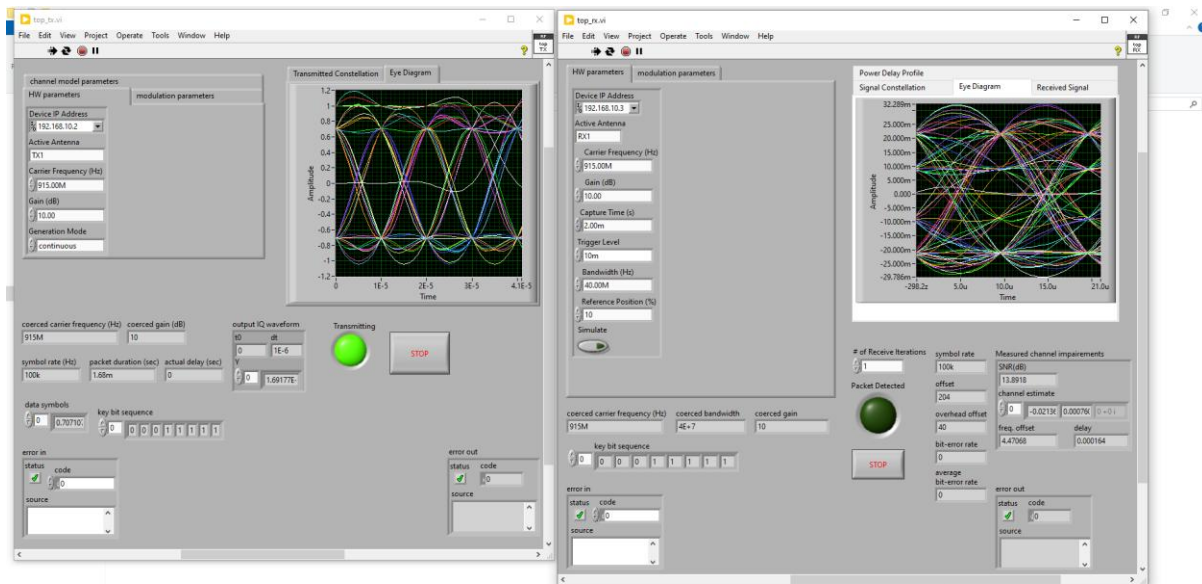


Figure 29:top_tx.vi & top_rx.vi(Eye diagram)

Questions and answers:

- **“MT Generate Filter Coefficients.vi” requires an input for the number of “pulse shaping samples per symbol”. This tells the VI the over sample factor being used to design the filter. When using this VI in “pulse_shaping.vi” where do you obtain this parameter from? Give any relevant cluster names and/or variable names if applicable.**

Ans: This Parameter can be obtained by TX oversampling factor cluster (derived from unbundle the modulation parameter

- **According to [1], impairments visible in the eyes diagram can occur at many places along the communication path (i.e. from source to sink). What are some of places where impairment visible in the eye diagram might occur (name at least three)?**

Ans: Pre-filtering in the transmission, through the frequency conversion and amplifier chain, propagation path, receiver front end, IF circuit, Baseband signal processing.

- **When Using inconsistency sampling rate at the transmitter and receiver, it is important to remember that the symbol rate ($1/T_s$) must remain constant.**

a) In terms of the sampling rates at the transmitter and receiver ($1/T_M$ and $1/T_N$ respectively), what is the relation between the oversample factor at the transmitter (M) and receiver (N)?

b) What can you say about type of number the ration of T_M/T_N (i.e. the ration of sampling rate of the receiver to that of the transmitter) must be?

Ans: a) $M=N$ They should be of same values.

b) Ration of T_M/T_N must be 1.

- **Using the pulse shaping parameter control on the front panels of top_tx.vi and top_rx.vi, vary the filter parameter (i.e. the Rolloff factor) of a root raised cosine pulse from 0 to1 in increments of 0.20. Observe how the eye diagram (on the front**

panel of top_rx.vi) changes as you vary this excess bandwidth parameter. a) Describe what happens to the opening of eye as you increase the excess bandwidth parameter from 0 to 1. b) Explain what impairment is being added or removed from the received signal as you vary this parameter.

Ans: a) As we increase the value of filter parameter in transmitter side from 0 to 1, the signal quality become more and low because of distortion, noise, and impairments. More explanation in below point.

b) As we increase these parameter amount of distortion, Time variation at zero crossing are significantly increases on the other hand SNR hence eye-opening area is reducing. All reason is responsible for low quality signal at receiving end.

- **Based on what you have learned from [1], and from varying the pulse shaping parameters, what is the relation between the excess bandwidth parameter of the root raised cosine pulse and inter-symbol interference? Does ISI increase or decrease with increasing α and why.**

Ans: The rolloff factor α , often expressed as a percentage of bandwidth, indicates the excess bandwidth of a signal. For instance, a 50% excess bandwidth corresponds to $\alpha=0.5$. To mitigate inter-symbol interference (ISI), it is beneficial to increase the value of α toward 1. A higher α results in quicker decay of the time-domain pulse tails, leading to a smoother spectral rolloff but requiring more bandwidth. Reducing the time-domain tails of the pulse shape minimizes ISI caused by sampling errors. Therefore, the rolloff factor α serves as a tradeoff between excess bandwidth and resistance to ISI resulting from sampling inaccuracies.

- **Discuss any problem you might have encountered in either part 1 or Part 2 of Lab 2 and how you overcome these obstacles.**

Ans: in pulse shaping I faced problem with my simulation , I over come this problem with consulting instructor.

- **Describe how the code you have developed in lab for modulate.vi and decode.vi might be extended or organized to be used with an arbitrary constellation (i.e. if you were given any arbitrary constellation C as a 1-D array of complex doubles in LabView, explain how you would modify your code for modulation.vi and decode.vi to use this new constellation). Please only explain details of your code that pertain to your implementation of algorithms for symbol mapping, energy normalization and maximization -likelihood detection. You do not have to construct the VI, only an explanation is needed.**

Ans: Input to the Modulator.vi is 8-bit unsigned integer. If the input signal is complex double.

- For QPSK: no need to convert the Unsigned integer into CDB values. May directly used to the o/p.

- For BPSK: need to eliminated the Imaginary part directly from the i/p signal Input to the Decoder.vi is Complex Double.

- **The background section of this lab stated that two main goals for pulse shaping are:**
7 • To reduce the energy of spectral sidelobes (i.e. excess bandwidth requirements). •

To reduce the time domain pulse tails which can lead to inter-Symbol interference. Are these conflicting or complimentary goals and why?

Ans: As, α refer to the excess Bandwidth. Larger the value of α the tails of the time-domain pulse decay quickly, however it requires more Bandwidth. On the other hands reducing the time-domain tails of the pulse shape reduce the amount of ISI.

SNR(db)	0	2	4	6	8	10
BPSK	0.1320	0.056	0.028	0.014	0.002	0.000
QPSK	0.178	0.138	0.102	0.040	0.016	0.006

